

BUSINESS REQUIREMENTS DOCUMENT

Global Gateway Traffic & Billing Governance

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Project Sponsor	[Insert Name/Department]	Version	2.0 (Global Gateway Scope)

1. Executive Summary

Our Apigee API Gateway currently hosts approximately **200–400 active API proxies**. These proxies are functioning correctly and already contain project-specific quota, Spike Arrest, validation, authentication and security controls. The business problem is now at the **platform level**: aggregate API traffic entering and leaving the gateway is higher than expected, increasing gateway-related consumption and cloud billing.

Because modifying 200–400 production proxies is inefficient and introduces regression risk, the proposed **Global Gateway Traffic & Billing Governance** solution will use an **Apigee Shared Flow attached through the environment-level Pre-Proxy Flow Hook**. The global layer will observe and classify traffic, read centrally managed KVM rules, protect gateway/backend capacity with targeted traffic controls, and provide a controlled path from monitoring to enforcement—while leaving existing proxy implementations unchanged.

Business Value: Establish a centralized, zero-touch governance layer for the existing proxy estate, improve attribution of excessive traffic, protect gateway/backend capacity, and provide a controlled mechanism to reduce unnecessary traffic and billing exposure without editing the 200–400 existing proxies.

2. Project Objectives

- **Gateway Traffic Visibility:** Measure aggregate traffic and attribute volume to individual proxies, consumers/applications and API products so the source of unexpected growth can be identified.
- **Zero-Touch Governance:** Introduce platform-level controls through a Shared Flow and Environment Flow Hook without editing, redeploying or redesigning the existing API proxies.
- **Cost & Capacity Control:** Reduce avoidable traffic and protect gateway/backend capacity through centrally managed, targeted controls rather than blindly applying one hard limit to every API.
- **Dynamic Management:** Allow platform operations to modify governance modes, thresholds, proxy rules and bypass entries through centralized KVM configuration.
- **Safe Rollout:** Support MONITOR → ALERT → TARGETED THROTTLE → ENFORCE stages, including an emergency bypass mechanism for critical services.



Operational rollout:

MONITOR → ALERT → TARGETED THROTTLE → ENFORCE
Global controls are introduced without editing existing proxies.

NOTE

Existing authentication, quota, Spike Arrest, validation and security remain unchanged.

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3. Functional Requirements

The following requirements define the centralized governance layer for the existing Apigee environment. The design intentionally separates **platform-level governance** from the existing project-specific policies already implemented inside the 200–400 API proxies.

Req ID	Feature Name	Description & Acceptance Criteria	Priority
FR-01	Pre-Proxy Flow Hook Integration	Description: Develop a centralized Apigee Shared Flow for gateway-level traffic governance and attach it to the environment's Pre-Proxy Flow Hook. Criteria: The Shared Flow executes automatically for incoming requests across the selected 200–400 existing proxies without modifying or redeploying those proxies.	Critical
FR-02	Gateway Traffic Intelligence	Description: Capture gateway-wide traffic context such as proxy, consumer/application, API product, HTTP method, status, latency and traffic volume for attribution and anomaly analysis. Criteria: Operations can identify which proxies and consumers are driving excessive traffic before enforcement is enabled.	Critical
FR-03	Centralized Gateway KVM Configuration	Description: Store global governance mode, proxy-level rules, thresholds and bypass/allowlist configuration in an encrypted KVM. Criteria: Operations can change governance settings without editing or redeploying the existing API proxies. Runtime changes follow the configured KVM cache/refresh behavior.	High
FR-04	Targeted Spike Protection	Description: Apply centrally managed Spike Arrest controls to selected proxies or traffic groups to protect gateway/backend capacity from abnormal bursts while preserving existing proxy-level policies. Criteria: Controls are conditional, dynamically configurable and do not replace application-specific quotas or security policies.	High
FR-05	Whitelist / Bypass Mechanism	Description: Provide a KVM-based allowlist for mission-critical proxies and approved traffic that must not be affected by global throttling. Criteria: Whitelisted entries bypass the applicable global enforcement rule while remaining subject to their existing proxy policies.	High
FR-06	Controlled Enforcement	Description: Support a staged governance mode: MONITOR, ALERT, TARGETED THROTTLE and ENFORCE. Criteria: The platform team can introduce controls gradually and use an emergency bypass/kill switch without changing the existing proxies.	High

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4. Non-Functional Requirements (NFRs)

Because the Shared Flow is attached to an environment-level Flow Hook, every applicable request can pass through the governance layer. Performance, high availability, backward compatibility and controlled rollout are therefore critical architectural constraints.

Area	Requirement
Performance & Latency	The global Pre-Proxy Shared Flow must add minimal latency. Target ≤15 ms for KVM/configuration evaluation and traffic-control logic under normal conditions, with performance validated under representative gateway load.
Zero-Regression Deployment	Deployment of the Shared Flow and Flow Hook must not alter existing proxy business logic, authentication, security validation, quota, Spike Arrest or response formatting except when a configured global governance rule intentionally blocks or throttles the request.
High Availability	Global traffic controls must use Apigee distributed runtime capabilities and be tested across message processors. The solution must not silently fail open when a governance decision is required.
Scalability	The architecture must support the existing 200–400 proxy estate and be load-tested against the measured gateway peak. Any target TPS must be validated rather than assumed; policy overhead and distributed counter behavior must be measured under load.
Security	The global layer must not weaken existing authentication or authorization. Consumer-specific enforcement should rely on trusted context established by the existing proxy/security model where available. Global governance must preserve existing security controls.

5. Monitoring & Proactive Alerting

- **Budget / Capacity Threshold Alerts:** Configure GCP/Apigee monitoring for agreed gateway consumption thresholds, with notifications to Cloud FinOps and the API Management team before the approved capacity or billing budget is approached.
- **Proxy-Level Attribution:** Analytics and operational telemetry must identify which proxies, consumers/applications and API products contribute the highest traffic, even when the global governance layer is in monitor or enforcement mode.
- **429 / Throttling Visibility:** Track HTTP 429 responses and global governance decisions separately from existing proxy-level quota/Spike Arrest violations so platform teams can distinguish application-level limits from platform-level controls.
- **Flow Hook Diagnostics:** Use targeted Apigee Trace sessions and latency measurements to isolate Shared Flow / Flow Hook overhead from existing proxy and target execution.
- **Governance Dashboard:** Provide visibility into total gateway traffic, traffic growth, top proxies, top consumers, throttled requests, bypassed critical proxies and governance mode.

Governance Principle: Existing proxy policies remain the source of truth for application-specific security and consumption requirements. The global layer provides platform-wide visibility and protection and should not duplicate or silently override those controls.

6. Approval

Cloud FinOps Approval	Signature & Date
Lead API Engineer Approval	Signature & Date